Validation Prevents False Confidence

A single high training score is not enough to trust a model.

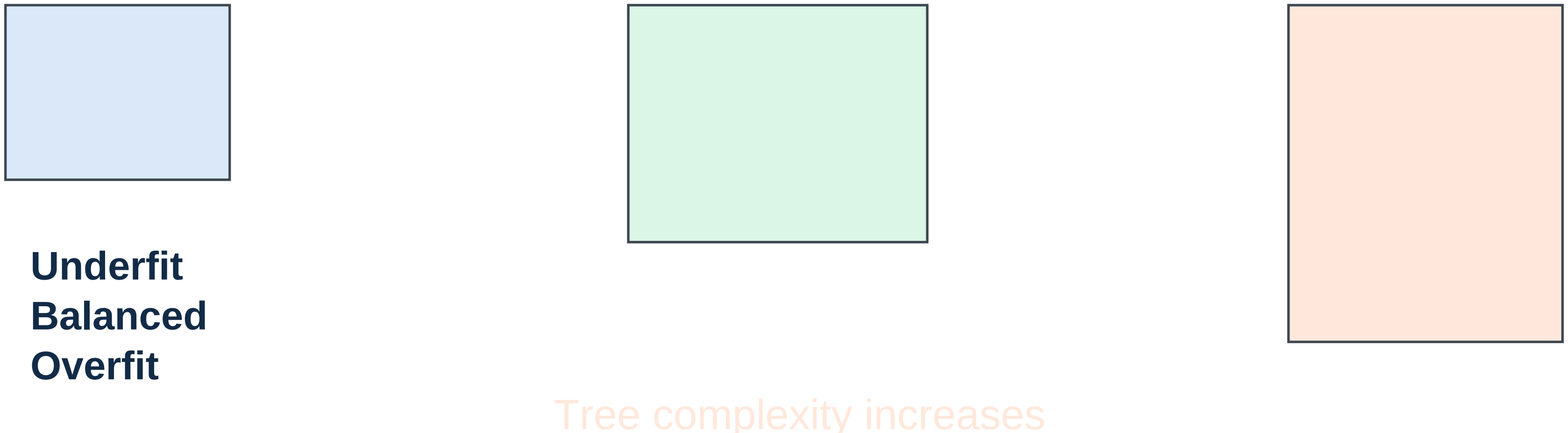
- Split the data into train and test sets before modeling.
- Use cross-validation to estimate performance across several folds.
- Track both train and validation/test error to detect overfitting.
- Evaluate with RMSE and R2 so error size and explained variance a

alize

Decision Trees Make Overfitting Easy to See

Tree depth and split rules control the bias-variance tradeoff.

- Very shallow trees underfit because they cannot capture enough structure.
- Unconstrained trees often memorize training data and fail on new rows.
- max_depth, min_samples_split, and min_samples_leaf reduce model complexity.
- Validation curves show where validation RMSE stops improving.



GridSearchCV Selects Tuned Settings

The final model is chosen using cross-validated validation performance, then checked once on the test set.

- Define a parameter grid for max_depth, min_samples_split, and min_samples_leaf.
- Fit GridSearchCV with 5-fold cross-validation and RMSE scoring.
- Compare baseline, overfit tree, and tuned tree using test RMSE and R2.
- Select the model that generalizes best, not the one with the lowest training error.